

Deploy a Flask + React Project to Hugging Face Spaces

Free, no credit card - full app in one Docker container - live *.hf.space URL

Live: <https://sachalchandio-sachal-portfolio.hf.space> (your real app - React site + Flask /api - served together from one free container; HF builds and runs your existing Dockerfile, nothing stripped).

1 - How it works

A Hugging Face Space is just a git repo that HF builds and runs. For a Docker Space, HF looks for a Dockerfile and a README.md whose top declares the SDK and port. Your repo's root Dockerfile is multi-stage:

```
# stage 1 - build the React app
FROM node:22-alpine AS web
COPY frontend/ . ; RUN npm ci && npm run build -> /web/dist

# stage 2 - Flask serves the built SPA + the API on one port
FROM python:3.12-slim
COPY backend/ . ; RUN pip install -r requirements.txt
COPY --from=web /web/dist ./webroot
CMD gunicorn --bind 0.0.0.0:${PORT:-8000} app:app
```

Flask serves webroot (the React build) for normal pages and answers /api/* itself - so the whole app is one origin, one container.

2 - Prerequisites (already in the repo)

- A working Dockerfile at the repo root (above).
- A README.md with Hugging Face front-matter so HF knows the SDK and port:

```
---
title: Sachal Chandio Portfolio
emoji: rocket
sdk: docker
app_port: 8000 # must match the port the container listens on
pinned: false
---
```

3 - The deploy steps

1. Create a free account at huggingface.co/join (email or GitHub/Google - no card).
2. Create a Write token at huggingface.co/settings/tokens -> New token -> type Write -> copy (hf_...).
3. Save the token into a gitignored file (here, .env as HF_TOKEN=hf_...) so it never appears in a command or in version control.
4. Run the deploy script (next section). It creates the Space, uploads the project, and HF starts the build automatically.
5. Wait for the build (~3-5 min the first time) and open the live URL.

4 - The deploy script (huggingface_hub)

```
pip install -U huggingface_hub
from huggingface_hub import HfApi
api = HfApi(token=TOKEN) # TOKEN read from .env
user = api.whoami()['name']
repo = f'{user}/sachal-portfolio'
api.create_repo(repo, repo_type='space', space_sdk='docker', exist_ok=True)
api.upload_file(path_in_repo='README.md', ...) # the front-matter above
api.upload_folder(folder_path='.', repo_id=repo, repo_type='space',
ignore_patterns=['**/node_modules/**', '**/dist/**', '.env', ...])
```

5 - Verify it's live

```
https://USER-sachal-portfolio.hf.space/ -> homepage + 3D loads
/off-duty /projects -> client routes work on refresh
/api/portfolio /api/off-duty /healthz -> JSON responds
contact form submit -> success (POST /api/contact)
```

6 - Updating after you make changes

Key idea: the Space is a git repo - any new commit or upload triggers an automatic rebuild and redeploy. The URL never changes. Pick whichever way fits:

Option A - re-run the upload (what we use): edit files locally, then run the same upload. `upload_folder` hashes files and only sends what changed, then HF rebuilds (~3-5 min).

```
from huggingface_hub import HfApi
api = HfApi(token=TOKEN) # from .env
api.upload_folder(folder_path='.', repo_id='USER/sachal-portfolio',
repo_type='space',
ignore_patterns=['**/node_modules/**', '**/dist/**', '.env', ...])
# -> commit pushed; Space rebuilds automatically
```

Option B - git push to the Space: set the remote once, then push on every change.

```
# one time:
git remote add space https://USER:hf_TOKEN@huggingface.co/spaces/USER/sachal-portfolio
# each update:
git add -A && git commit -m 'update' && git push space main
```

Option C - edit in the web UI: open the Space -> Files -> edit a file -> Commit. Best for one-line tweaks.

After any of these, watch progress on the Space's Logs / App tab, or poll `HfApi().get_space_runtime(repo).stage` until it reads `RUNNING`. Example: dropping `frontend/public/profile.jpg` then re-running Option A puts your photo live on the next rebuild.

7 - Notes & limitations

- Free tier: 2 vCPU / 16 GB for the build, a small always-free CPU at runtime. No card.
 - The Space sleeps after a stretch of inactivity and wakes on the next visit (a few seconds). Fine for a portfolio.
 - First build takes a few minutes; later updates rebuild the same way (cached layers make them faster).
 - `app_port` in README must equal the port the container listens on (8000 here). Mismatch = builds but page won't load.
 - Revoke the write token after deploying if it was ever shown on screen - huggingface.co/settings/tokens -> delete.
 - The Space and its code are public by default (set private at create time if needed) - fine for a portfolio.
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